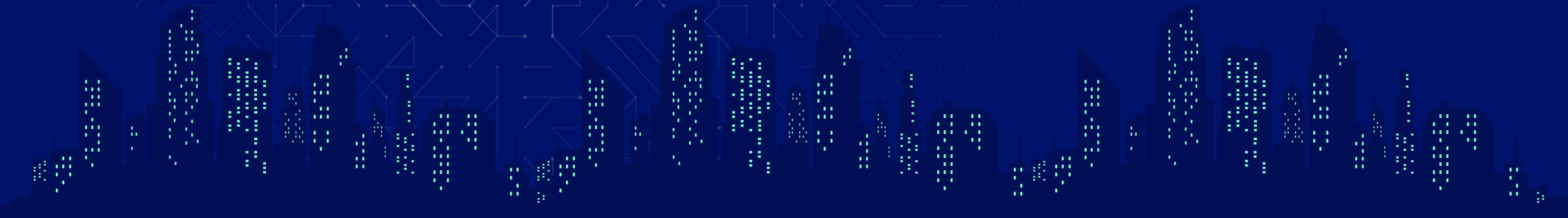
TECHORAMA

Secrets to Next Level Mobile Development with Xamarin

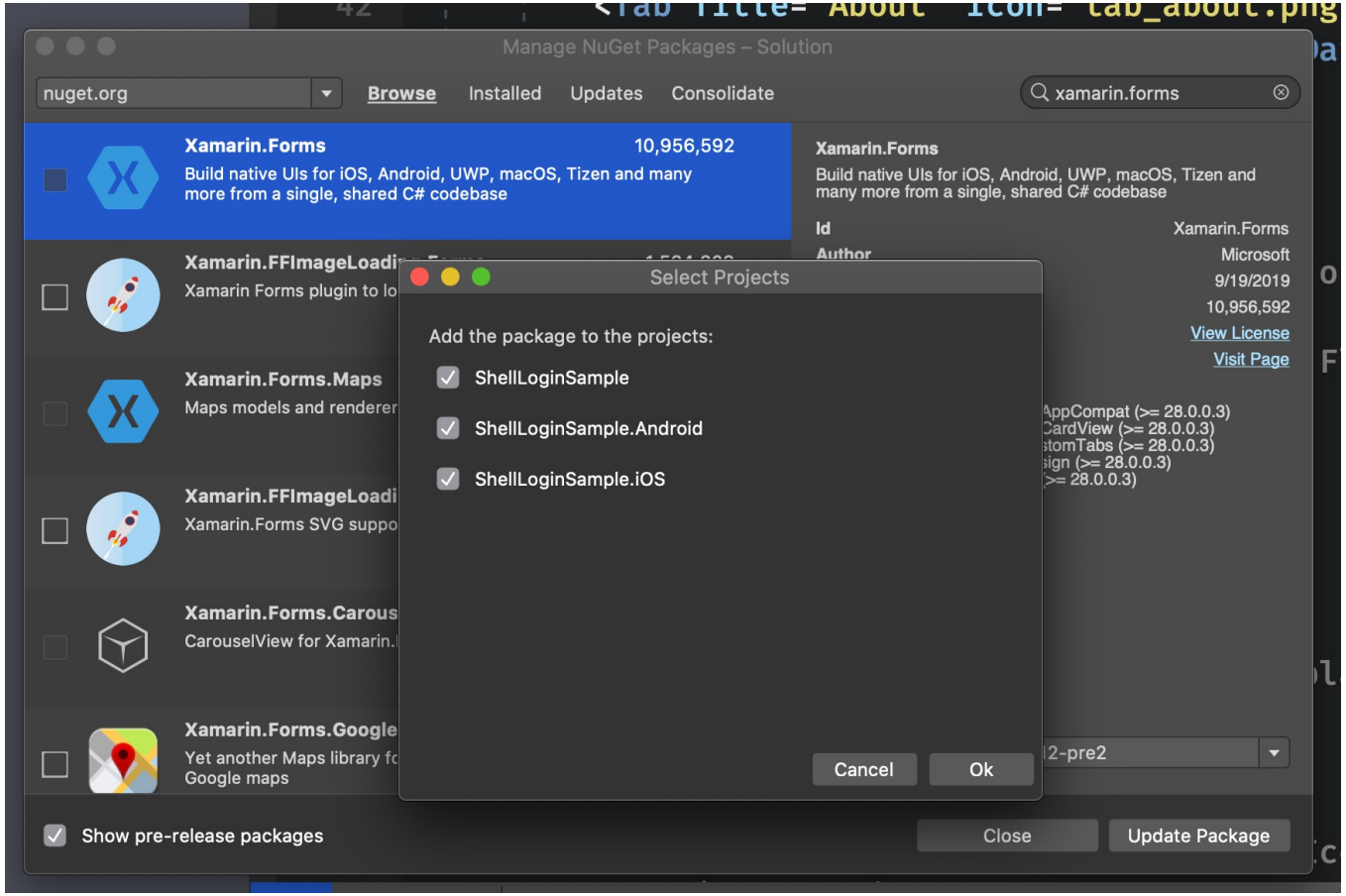
David Ortinau

Principal Program Manager, Microsoft

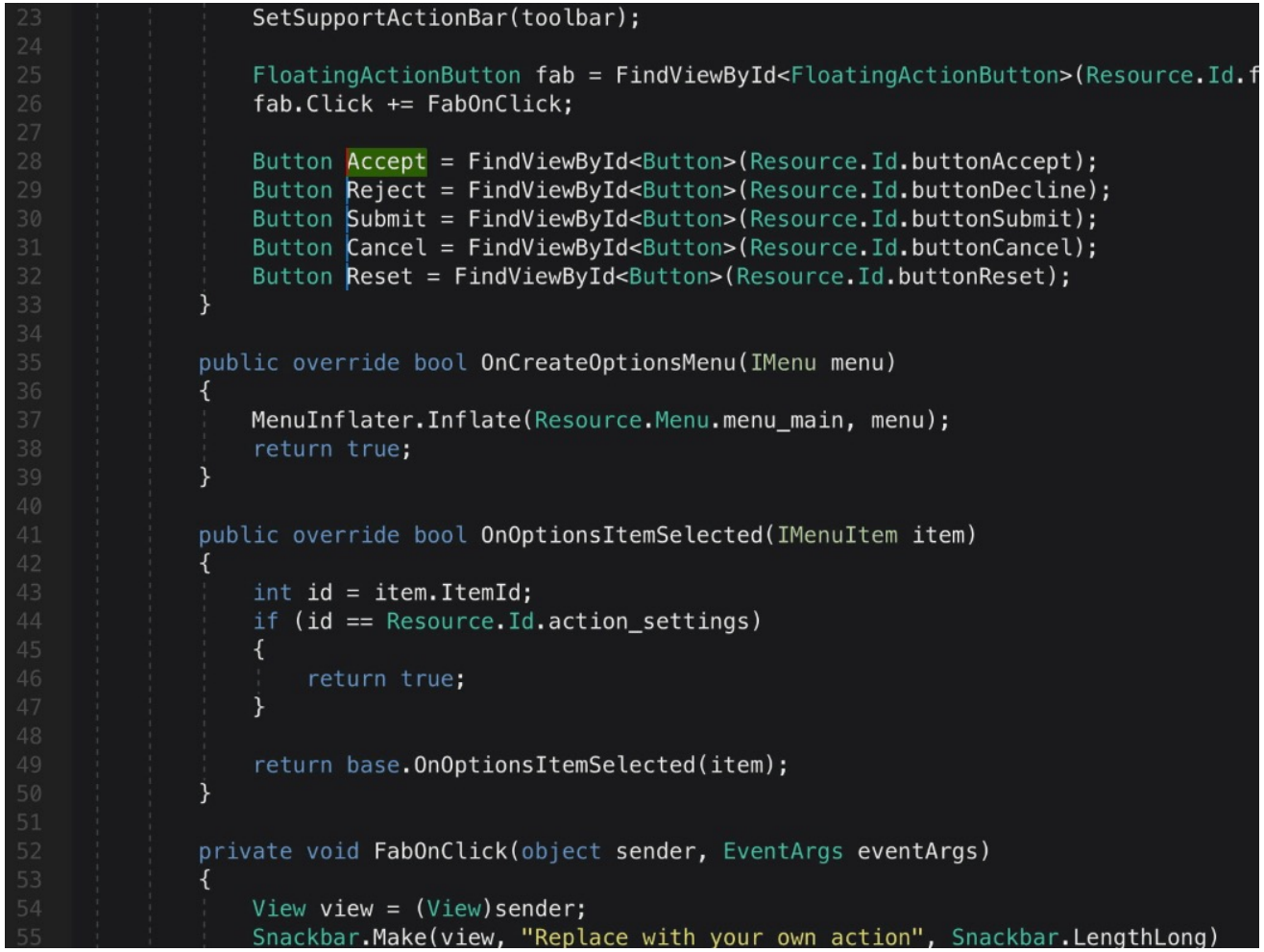


Productivity



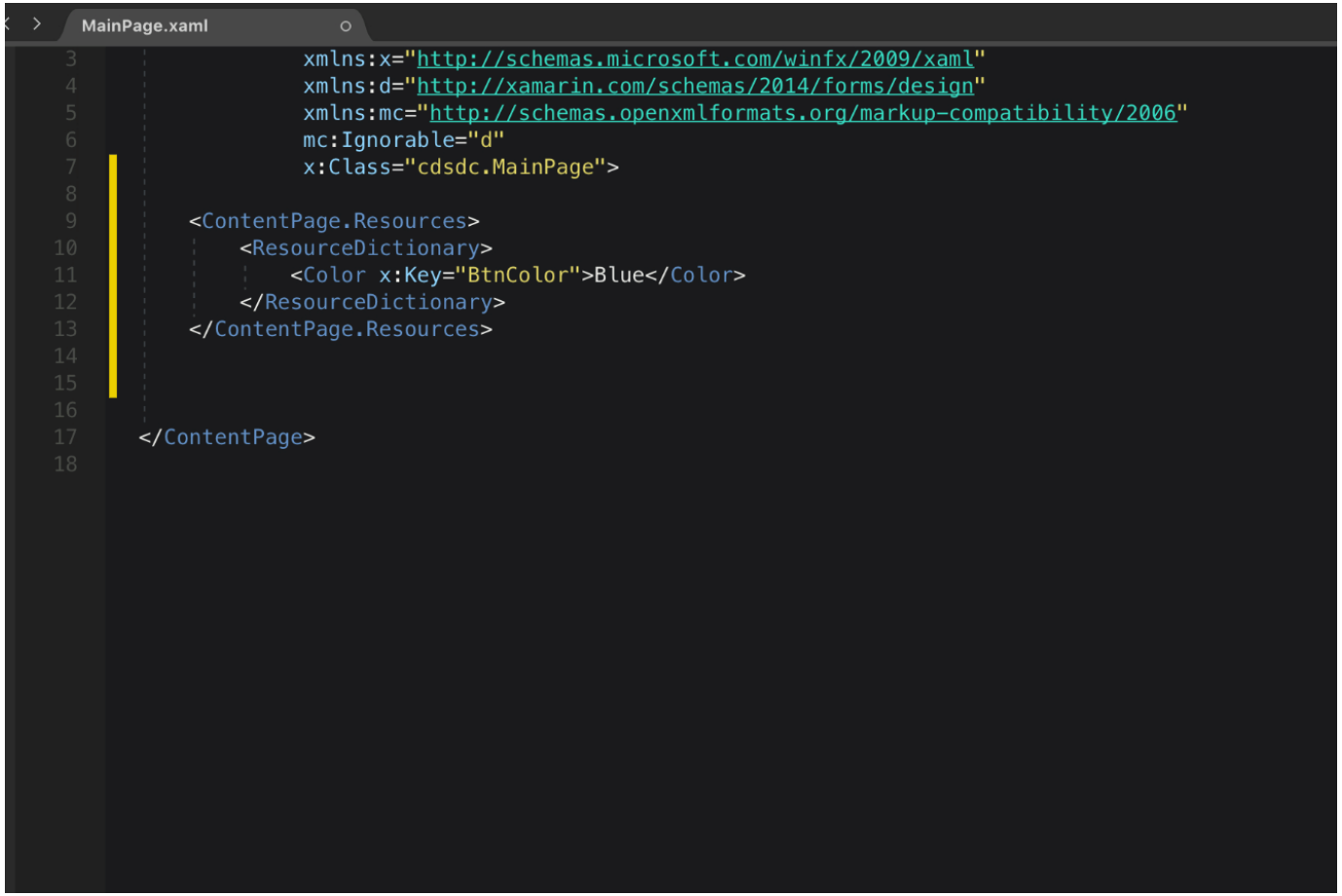


Solution Package Manager



```
23              SetSupportActionBar(toolbar);
24
25              FloatingActionButton fab = FindViewById<FloatingActionButton>(Resource.Id.f
26              fab.Click += FabOnClick;
27
28              Button Accept = FindViewById<Button>(Resource.Id.buttonAccept);
29              Button Reject = FindViewById<Button>(Resource.Id.buttonDecline);
30              Button Submit = FindViewById<Button>(Resource.Id.buttonSubmit);
31              Button Cancel = FindViewById<Button>(Resource.Id.buttonCancel);
32              Button Reset = FindViewById<Button>(Resource.Id.buttonReset);
33          }
34
35          public override bool OnCreateOptionsMenu(IMenu menu)
36          {
37              MenuInflater.Inflate(Resource.Menu.menu_main, menu);
38              return true;
39          }
40
41          public override bool OnOptionsItemSelected(IMenuItem item)
42          {
43              int id = item.ItemId;
44              if (id == Resource.Id.action_settings)
45              {
46                  return true;
47              }
48
49              return base.OnOptionsItemSelected(item);
50          }
51
52          private void FabOnClick(object sender, EventArgs eventArgs)
53          {
54              View view = (View)sender;
55              Snackbar.Make(view, "Replace with your own action", Snackbar.LengthLong)
```

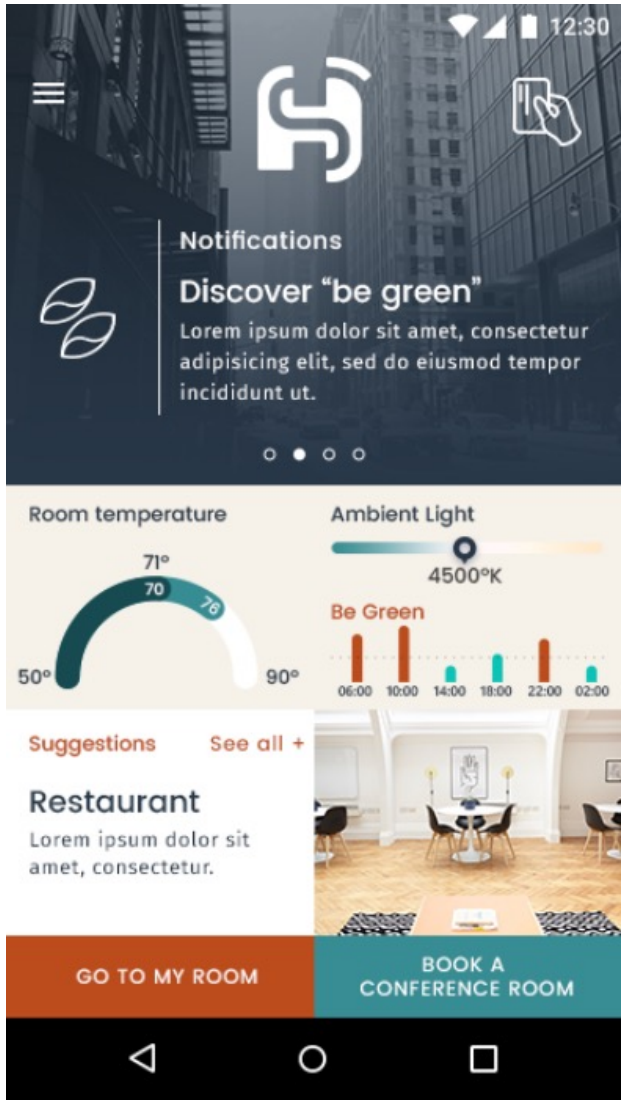
Optimized code editor - Mac



XAML IntelliSense - Mac

Faster Build & Deploys

Step	VS 2017 (15.8)	VS 2019 (16.0)	Delta
First Build	01:04.20	00:50.13	-21.95%
Incremental Build (XAML Change)	00:10.62	00:07.47	-29.66%
Deploy (XAML Change)	00:09.03	00:04.44	-50.83%



XAML Hot Reload

- Built into Visual Studio
- 16.3 / 8.3 – disabled by default
- 16.4 / 8.4 - enabled by default
- Works over debugger
  - Breakpoint
  - Injects Hot Reload agent
- Don't Link
- Reloads the entire page
- Calls OnAppearing

- What's Next
  - Partial reloading
  - C#
  - State restoration
  - Non-Forms scenarios

Xamarin Hot Restart (Private Preview)

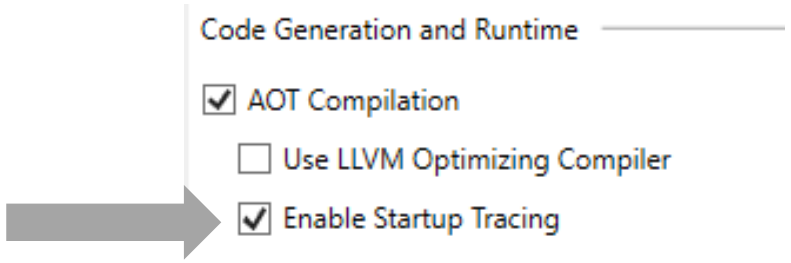
https://aka.ms/xamarin-hotrestart

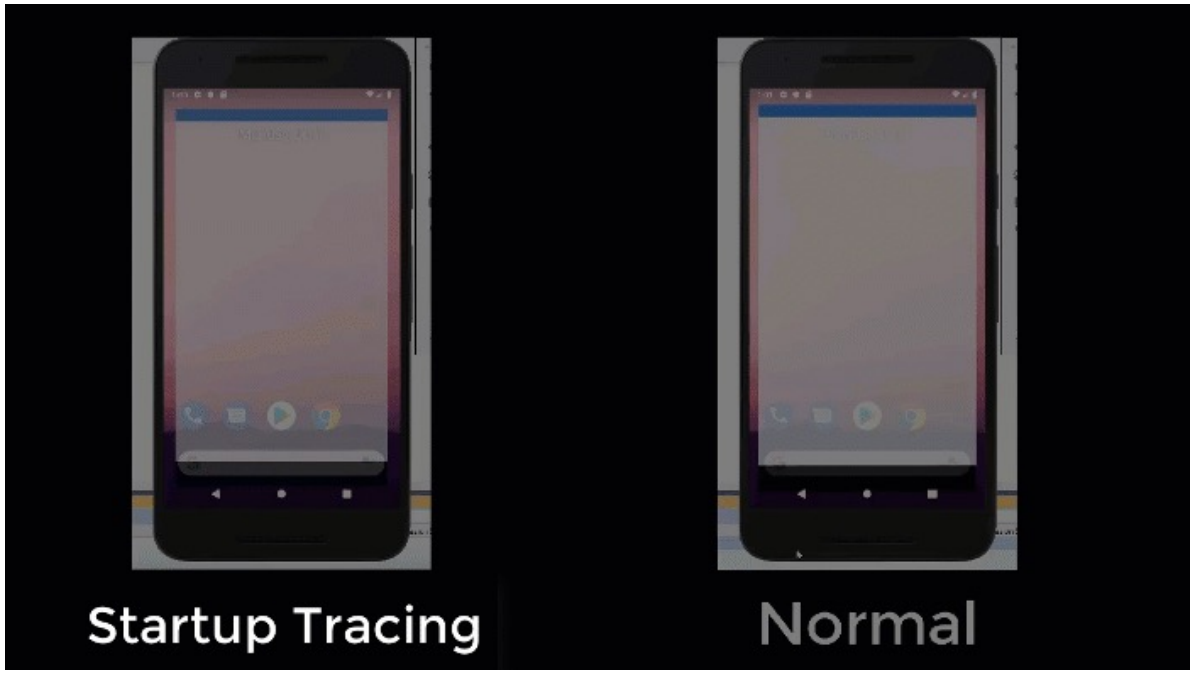
Platform Love



Android Startup Tracing

Type	Startup Speed	APK Size





Android App Bundles

Specific Languages

Proper Resource Screen Densities

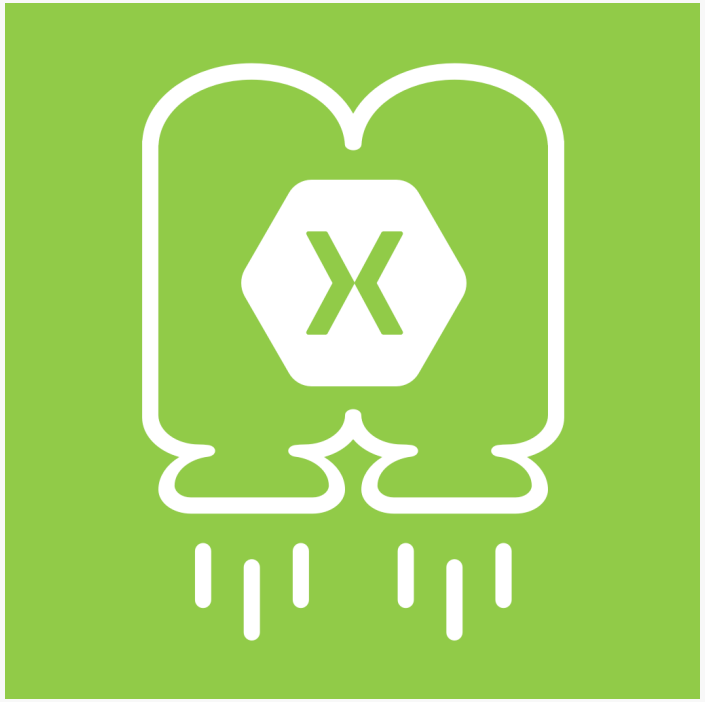
Best Match ABIs





What is AndroidX?

- The successor to the Android Support Library(created 2011).
- Known as “**Android** e**X**tensions”
  - Not to be confused with Android 10.
- Major Improvements such as:
  - Coherent package naming. No more v4, v7, etc!
  - Each package is separately maintained.
  - Features added as they pertain to each package.
- Available on NuGet.



AndroidX Packages

- Your favorite packages such as:
  - Fragment
  - AppCompat
  - RecyclerView
  - ConstraintLayout
  - CardView
  - VectorDrawable
  - WorkManager
  - CameraX
  - ViewPager
  - Wear
  - And many more!



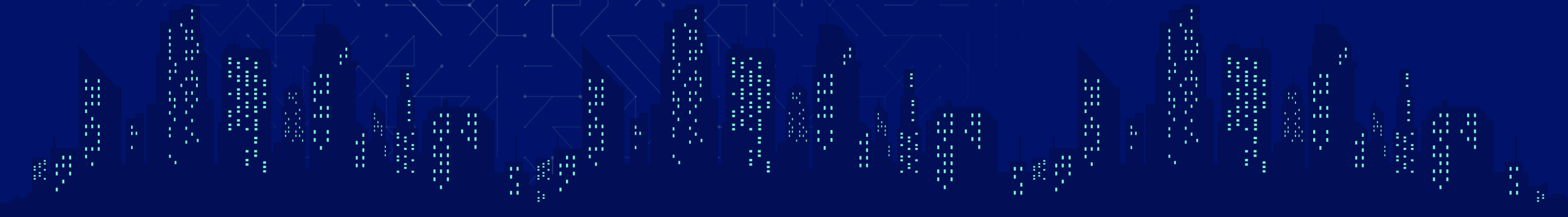
- Now come in new Jetpack collections!
  - Foundation
    - AppCompat
    - Wear
  - Architecture
    - WorkManager
    - ViewModel
    - Room
  - Behavior
    - CameraX
    - Notifications
  - UI
    - Fragment
    - Layout

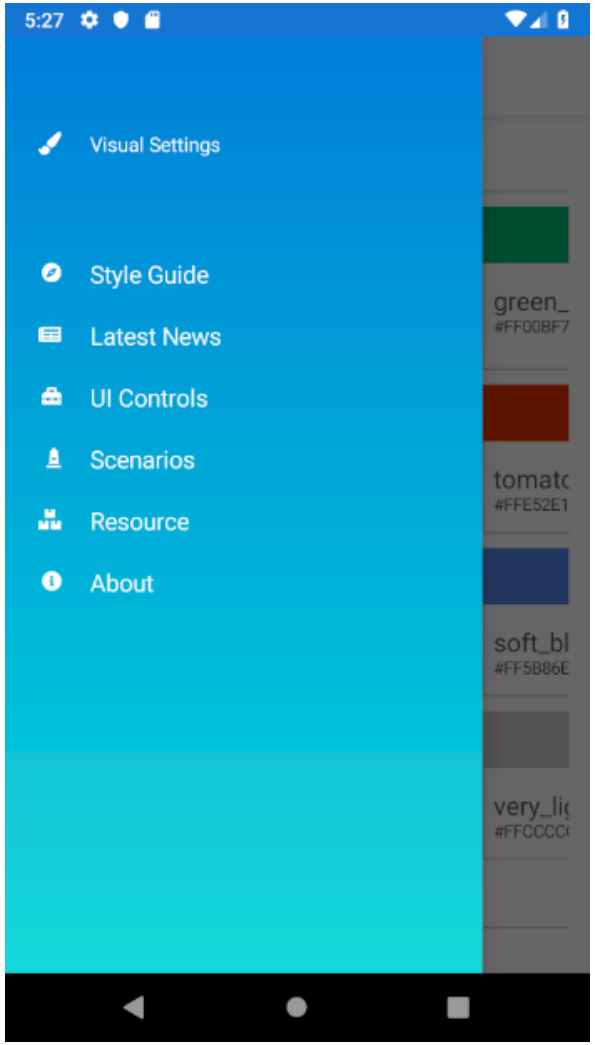
AndroidX Migration

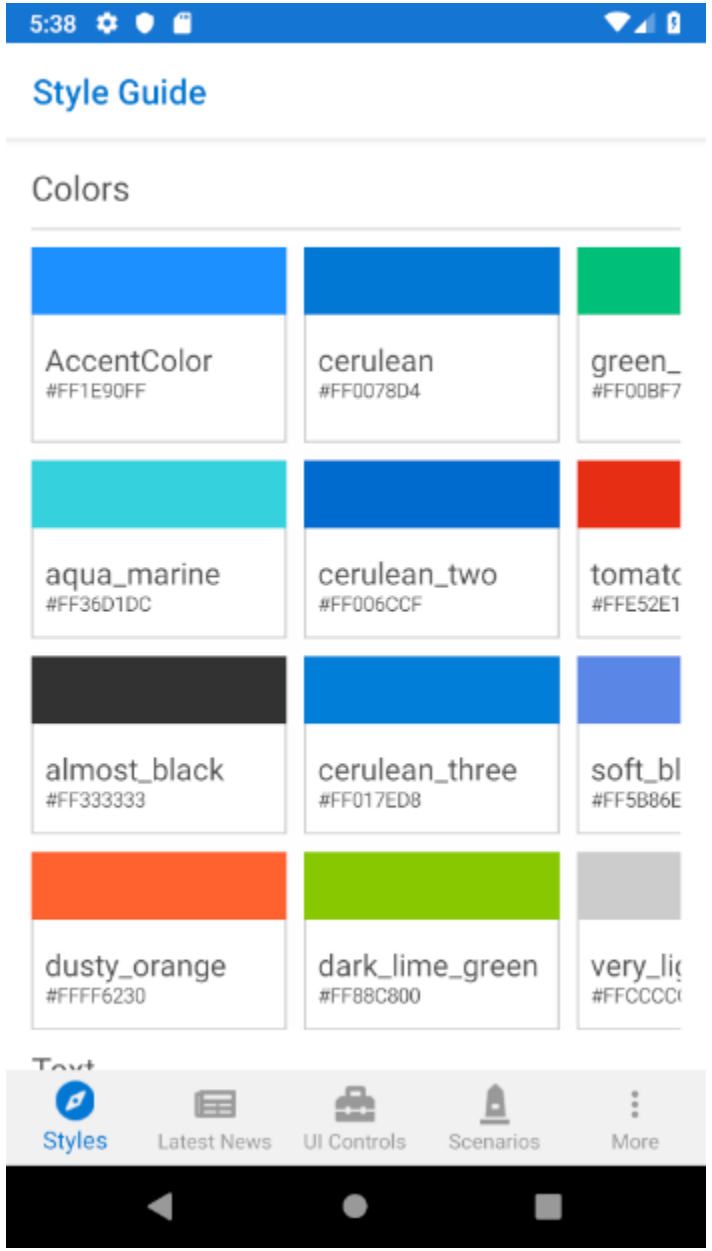
- Migration Tooling from the Android Support Library -> AndroidX.
  - Migration Package available on NuGet.
  - IDE One-Time Migration Tooling Available in 16.4/8.4 Previews.

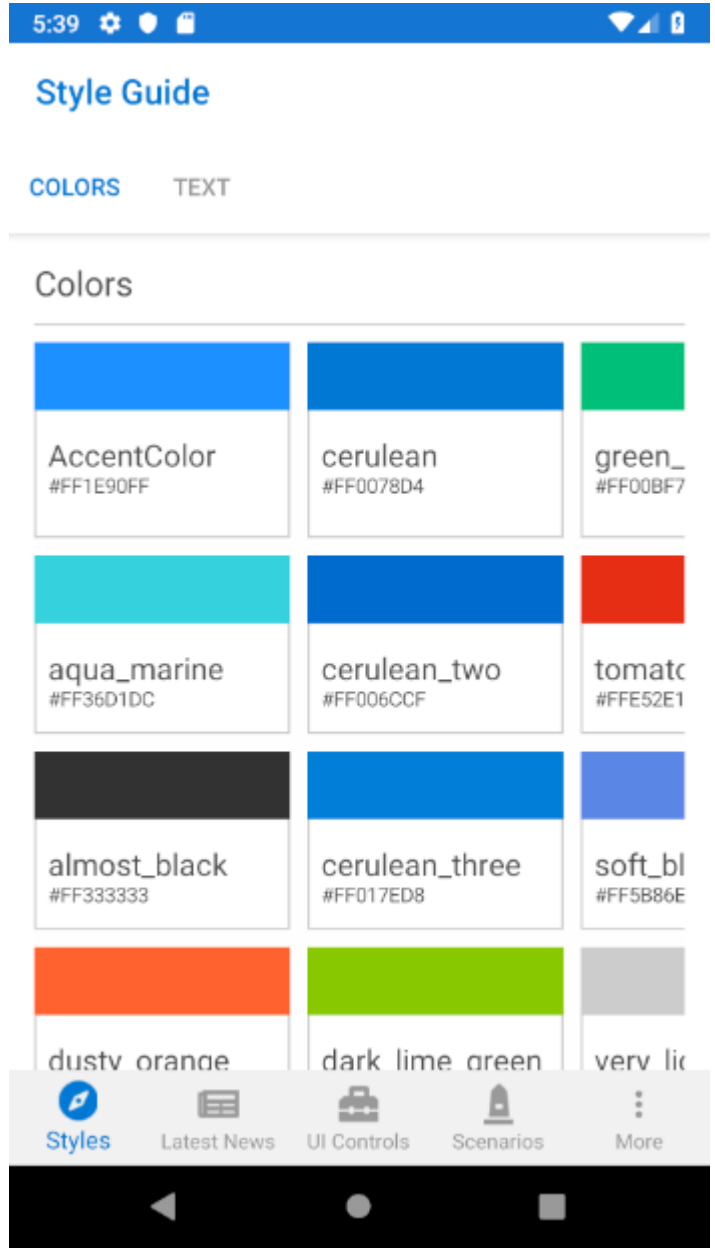


Use Shell

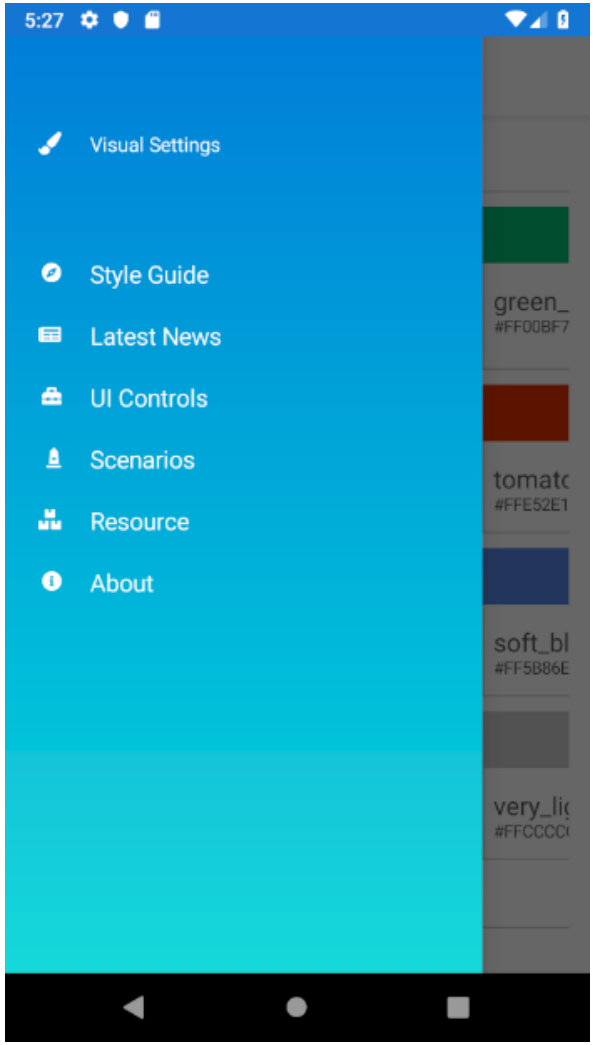



5:27
Visual Settings
Style Guide
Latest News
UI Controls
Scenarios
Resource
About


5:38
Style Guide
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
dusty_orange
#FFFF6230
dark_lime_green
#FF88C800
Styles
Latest News
UI Controls
Scenarios
More

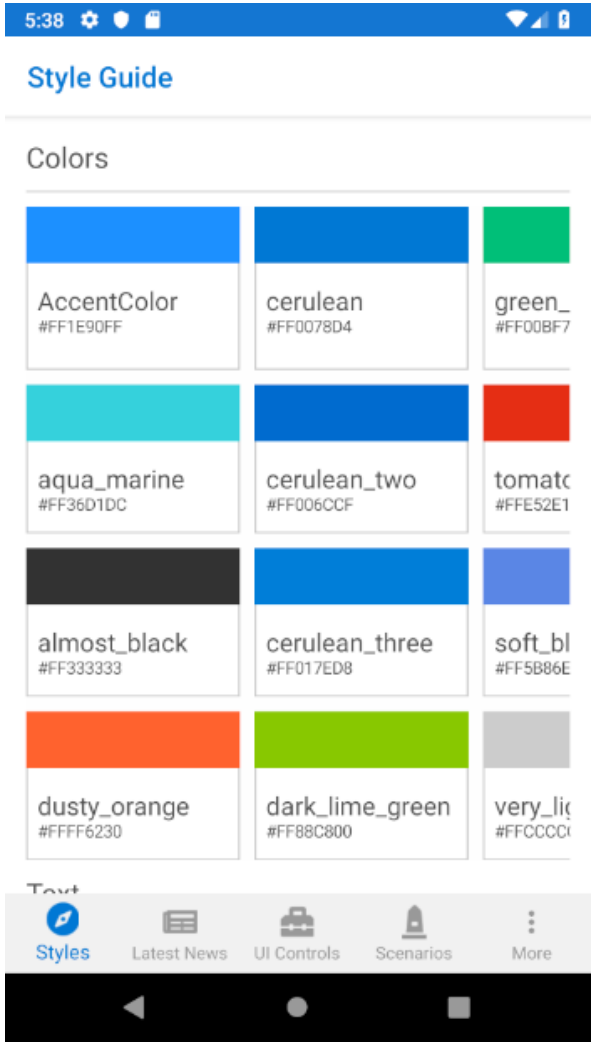

5:39
Style Guide
COLORS
TEXT
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
Styles
Latest News
UI Controls
Scenarios
More

Shell



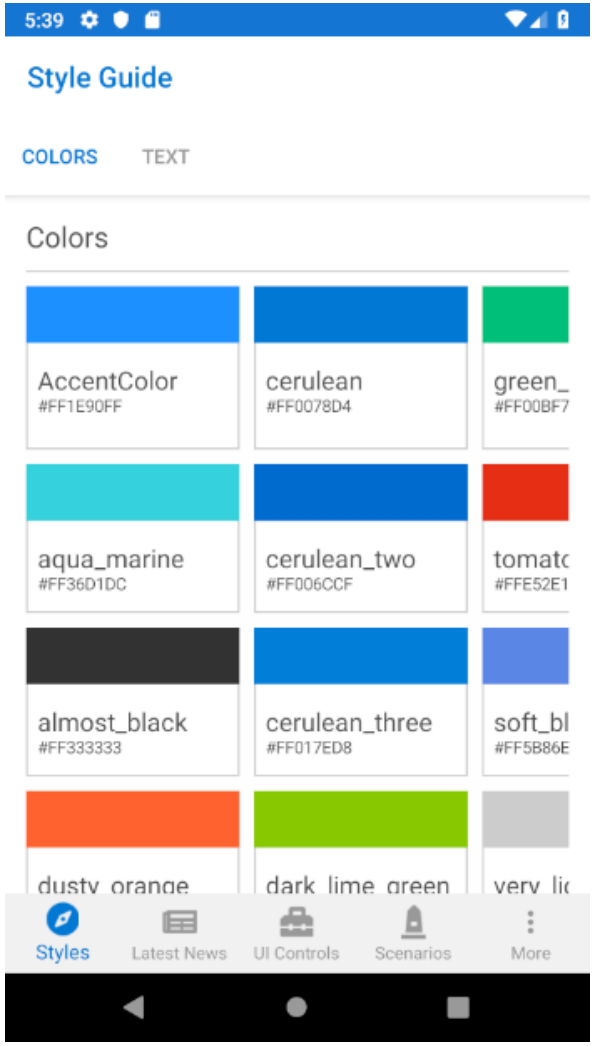

```
<Shell>
    <FlyoutItem
        Title="Style Guide"
        Icon="Compass.png">
        <ShellContent
            ContentTemplate="{DataTemplate p:StyleGuidePage}"/>
    </FlyoutItem>
</Shell>
```

Shell




```
<Shell>
    <TabBar>
        <Tab
            Title="Styles"
            Icon="Compass.png">
            <ShellContent
                ContentTemplate="{DataTemplate
                                  p:StyleGuidePage}"/>
        </Tab>
    </TabBar>
</Shell>
```

Shell




```
<Shell>
    <TabBar>
        <Tab
            Title="Style Guide"
            Icon="Compass.png">
            <ShellContent
                Title="Colors"
                ContentTemplate="{DataTemplate p:ColorsPage}"/>
            <ShellContent
                Title="Text"
                ContentTemplate="{DataTemplate p:TextPage}"/>
        </Tab>
    </TabBar>
</Shell>
```

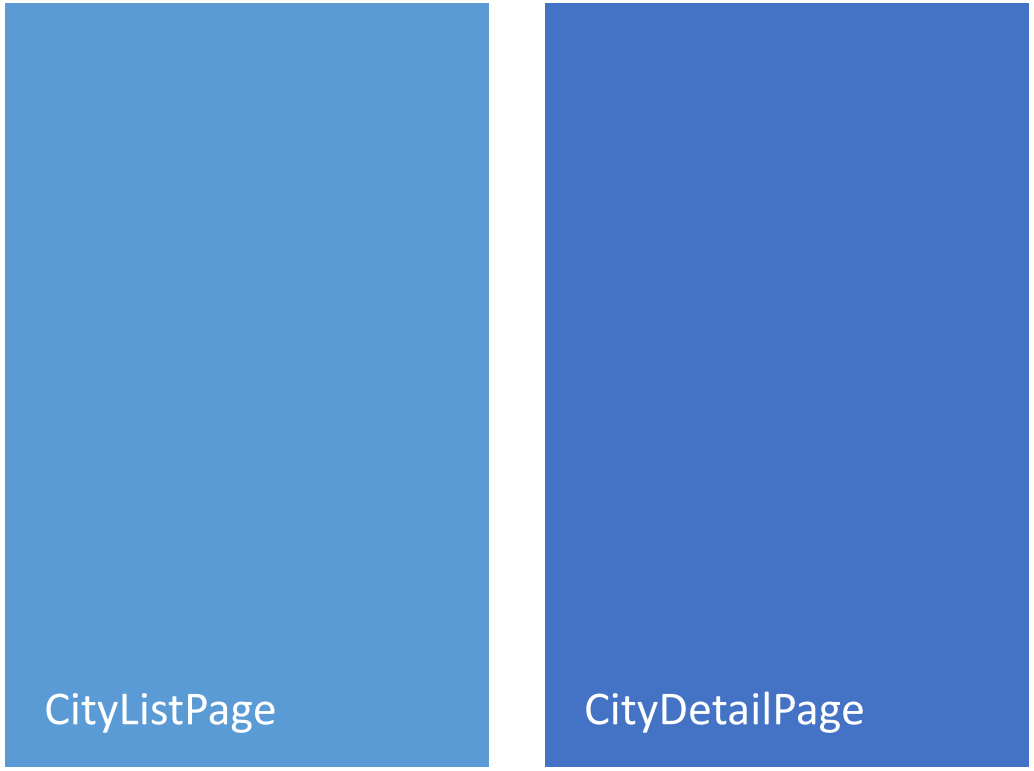
Shell




```
Shell.Current
       .GoToAsync($"detail?id={city.id}");
```



Shell Navigation Service




```
[QueryProperty("CityId", "id")]
public partial class CityDetailPage :
ContentPage
{
    public CityDetailPage()
    {
        InitializeComponent();
    }

    public string CityId
    {
        set
        {
            // do something with the Id
        }
    }
}
```

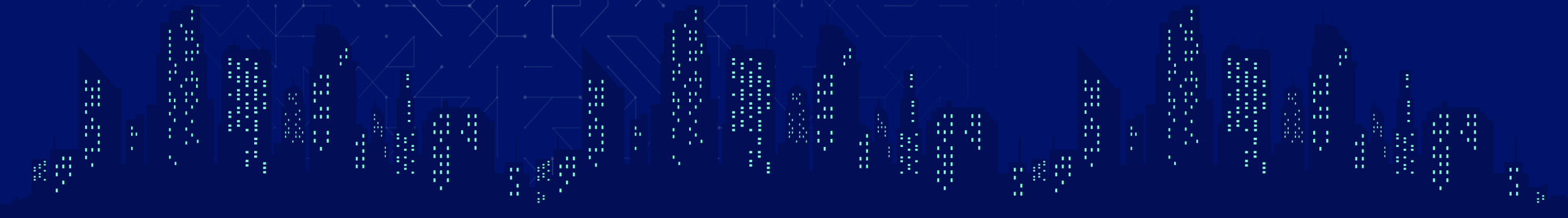
Shell Navigation Service

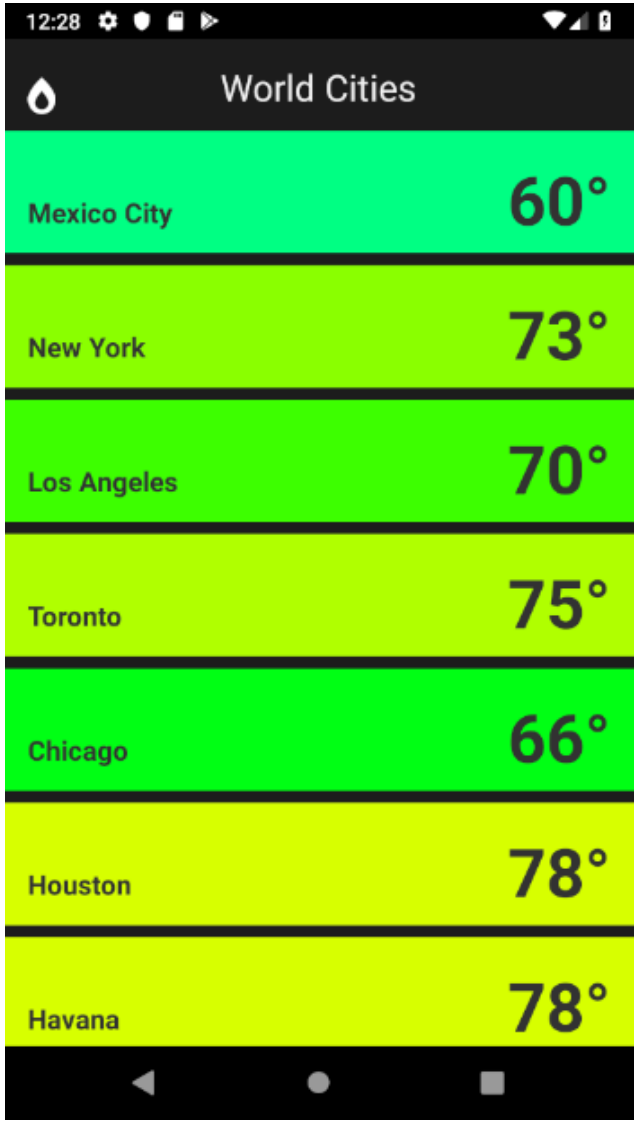
```
global::Xamarin.Forms.Forms.SetFlags("Shell_UWP_Experimental");
```

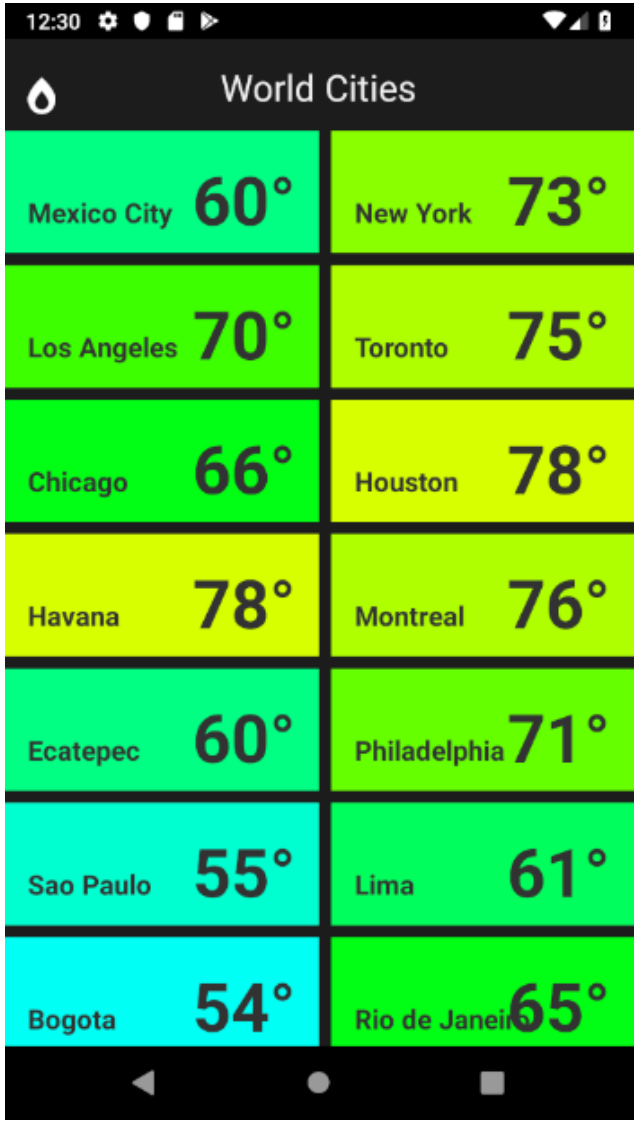
- Basic implementation
- Plenty of glitches
- Please file issues and pull requests
- Please file pull requests and issues

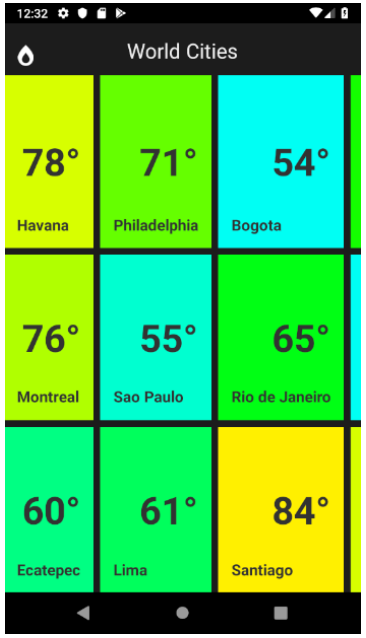
Shell for UWP (Preview)

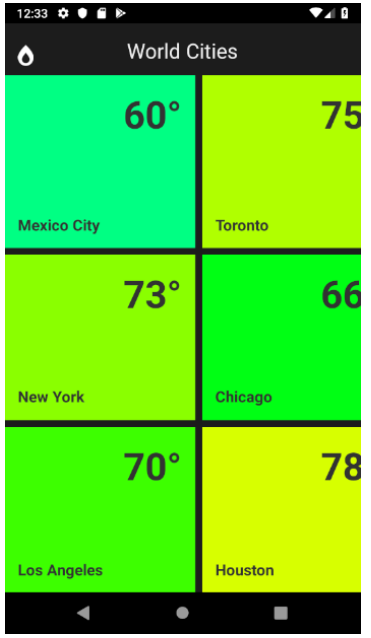
Xamarin.Forms 4.3 Features

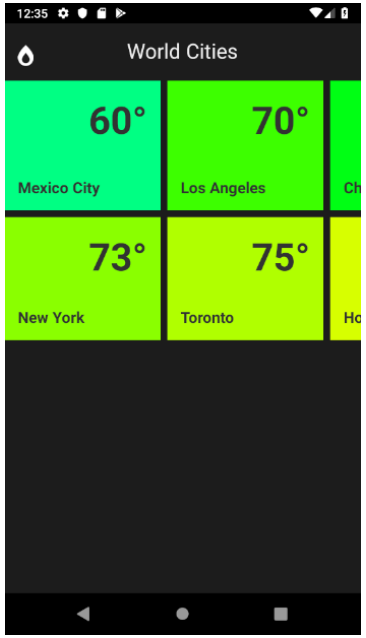


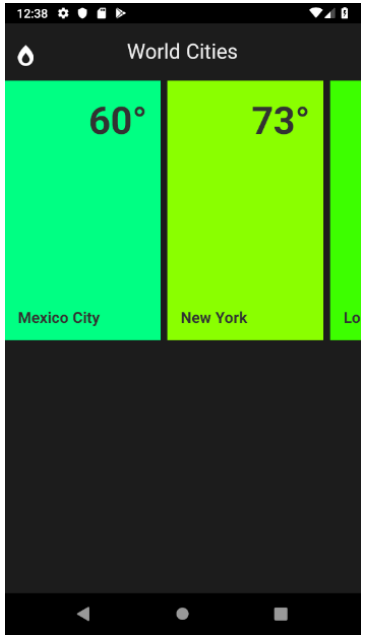








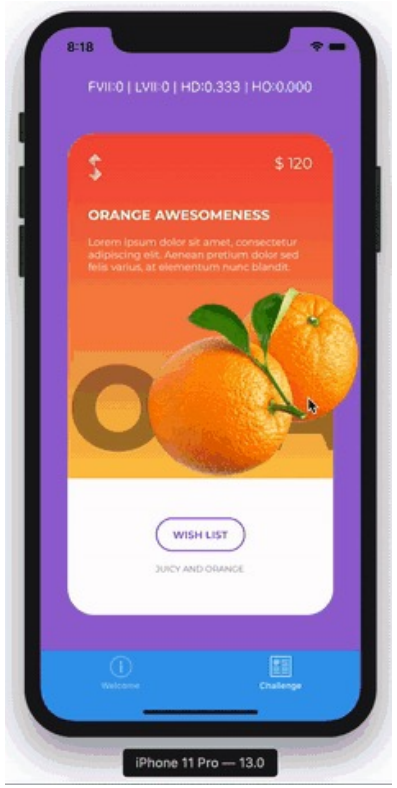




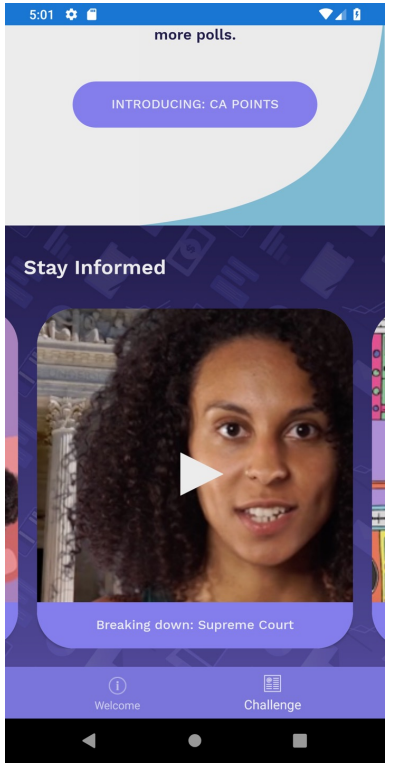
CollectionView

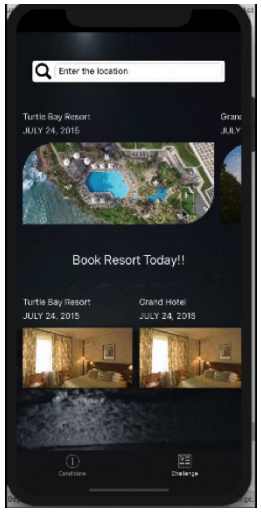


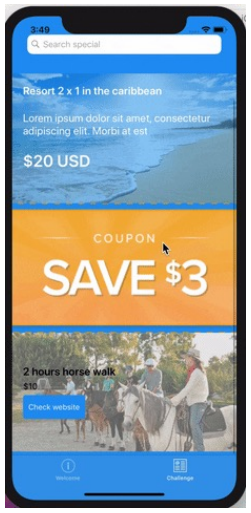
CarouselView

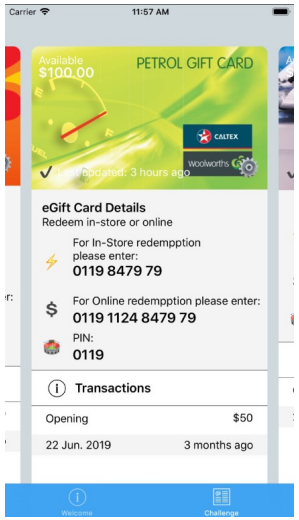


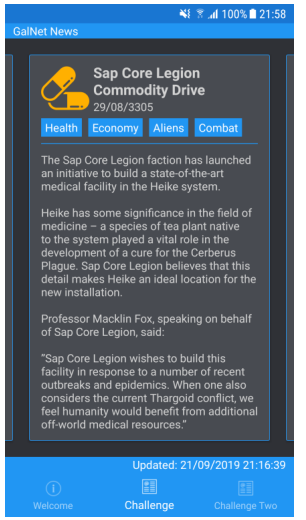


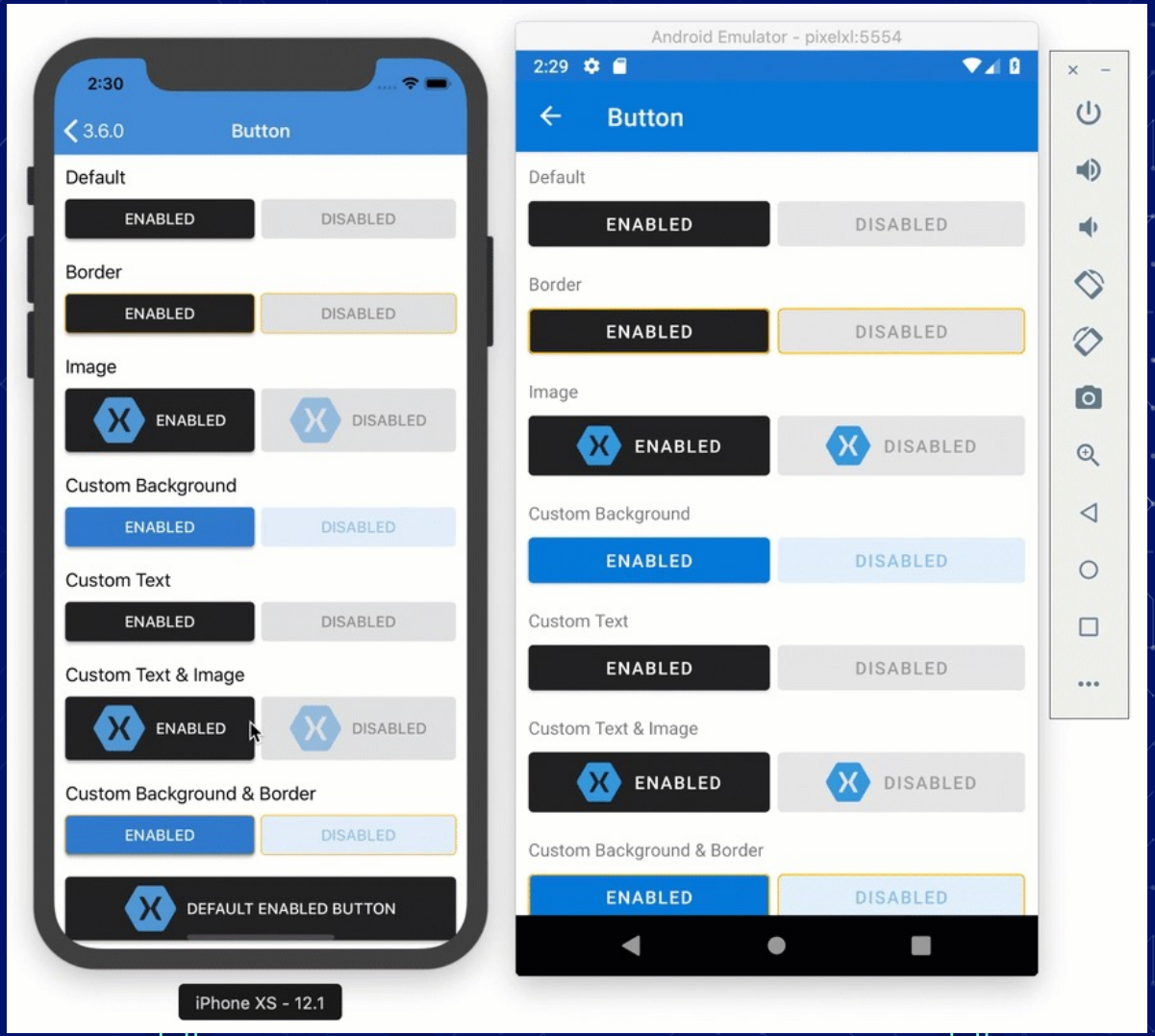












- Sets of renderers
- Adhering to a specific design system
- Beginning with Material Design

  - ActivityIndicator
  - Button
  - Editor
  - Entry
  - Frame
  - Picker
  - ProgressBar
  - Slider
  - Stepper

Visual

```
<Button Text="Consistent Design"
        Visual="Material"/>
```

Thank You!
Questions?

